

CROSS: A Mixture-of-Expert Reinforcement Learning Framework for Generalizable Large-Scale Traffic Signal Control

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Why:

- Adaptive traffic signal control (ATSC) adjusts signal phases based on real-time traffic observations to reduce congestion and improve traffic flow.
- In this project, we focus on networks with heterogeneous intersection layouts and traffic demands. A single shared RL policy may struggle to handle this diversity. We propose CROSS, a Mixture-of-Experts RL framework that learns specialized control strategies for different traffic patterns.

Conventional Algorithms

- Fixed-Time control follows a predetermined phase sequence and schedule, while Max-Pressure selects phases based on queue differences between incoming and outgoing lanes. These methods provide useful baselines, but their predefined control rules do not learn specialized strategies from experience.

Problem Statement:

- We consider a network of signalized intersections with different numbers of lanes, movements, and phases. Each intersection has its own agent, which selects a signal phase at each control step to reduce queues and improve traffic flow.
- Each agent observes only local traffic conditions, including queue lengths, moving-vehicle counts, and movement activation status. Phase configurations and topology information are also provided so the same network architecture can handle different intersection layouts.

RL Cast:

- State space:
 - o Five values per traffic movement: activation status, incoming/outgoing queue lengths, and incoming/outgoing moving-vehicle counts.
 - o Phase configurations and intersection topology features, including lane counts and speed limits.
- Action space: Select one collision-free phase from the intersection's available phases, applied for a fixed control interval.
- Reward structure: The negative sum of incoming-lane queue lengths, measured within a 50 m sensing range.
- Neural Network Structure: An MLP followed by a GRU encodes traffic histories, while another MLP encodes phase information; cross-attention fuses their features. A predictive clustering module learns traffic-pattern embeddings, which are concatenated with the fused features to route MLP experts. Their weighted outputs feed linear policy and value heads.

RL algorithms and envisioned results:

- We will use PPO with actor and critic parameters shared across intersections. Prediction, clustering, and routing objectives will be optimized jointly with the RL objectives.
- We will compare CROSS with GESA, Unicorn, Fixed-Time, and Max-Pressure using average queue length, speed, trip completion rate, and trip delay. We expect expert specialization to be particularly useful under heterogeneous traffic conditions.
- Experiments will use SUMO, with joint training on Grid 4x4, Arterial 4x4, and Grid 5x5 networks. We will evaluate zero-shot transfer and few-shot fine-tuning on real-world-derived networks to assess generalization.